A Cluster-Based Route Design of Multi-Capacity Vehicle in Large-Scale Demand Responsive Transport Service for the Disabled

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ABSTRACT

In large urban areas, many demand responsive transport (DRT) services dispatch vehicles on a first-come-first-serve basis for requests except for advance requests. Demand exceeding supply increases the share of requests not scheduled in large urban areas. If a DRT system can find a properly fixed route of a multi-capacity vehicle based on a spatiotemporal demand pattern, the efficiency of the DRT system will increase. This study aims to develop a method to design a cluster-based route for a multi-capacity vehicle in a large-scale network using historical data. The methodology of this study consists of four steps: 1) construct initial routes from requests; 2) calculate the similarity between two initial routes; 3) use a clustering process (density-based spatial clustering of applications with noise) to find spatiotemporally similar routes; 4) design a cluster-based route for a multi-capacity vehicle. This study evaluated the performance of a cluster-based route in a toy network and real large-scale network. In the toy network, an average waiting time of users on a cluster-based route decreased to 18% of users of a current service; further, average waiting time of users on the exact solution of a current service decreased to 9% of users of a current service. In the large-scale network, an average waiting time of users on a cluster-based route decreased to 20% of users of a current service. This study verifies that the efficiency of DRT service for the disabled increases by substituting a fixed route of a multi-capacity vehicle with a spatiotemporally similar route.

Keywords: Demand Responsive Transport Service, The Disabled, Route Design, Clustering Algorithm, Density-Based Spatial Clustering of Applications with Noise

INTRODUCTION

Mobility services for the elderly and the disabled began in the 1970s and 1980s around the world (*1*). Demand responsive transport (DRT) provides door-to-door service to passengers with impaired mobility, such as the elderly and the disabled. DRT, which provides transport on demand from passengers using fleets of vehicles scheduled to pick up and drop off people by their needs, has a flexible itinerary and route compared with that of traditional public transport (*2*). DRT services for the disabled dispatch vehicles operating on an optimal route for certain advanced requests and dispatch vehicles on a first-come-first-serve (FCFS) basis for other requests. Demand for DRT for the disabled has increased since laws ensuring the rights of the socially vulnerable have been established.

The Americans with Disabilities Act (ADA) was enacted in 1990 to guarantee mobility of the elderly and disabled in the United States (US) (*3*). Many DRT services designed to meet travel needs of persons whose mobility is limited are mandated by federal regulations that execute the transportation provisions of the ADA of 1990 (*3*). Access-A-Ride (NewYork City) and MetroAccess (Washington, D.C.) are examples of DRT services operated by the ADA of 1990. According to APTA, the ridership of DRT has increased about three times from 1991 to 2019 (*4*). In the United States, the ridership of DRT in the first quarter of 2019 was 51,994,000, a 2.63% increase from the previous year's first-quarter ridership of 50,661,000. The rate of increase in the ridership of DRT was the highest among surveyed modes of public transit (bus, regional rail, etc.) (*4*). The operating cost per rider of DRT is $25, which is the highest among modes of public transit (*4*). NewYork City's Access-A-Ride, for example, has seen a rise from 7% to 14% in the share of requests not scheduled from 2004 to 2015 (*5*).

In South Korea, the Act on Promotion of the Transportation Convenience of Mobility Disadvantaged Persons was enacted in 2005. Mobility services for the disabled and the elderly are compulsory to operate as a DRT system by local governments. In Seoul, the ridership of DRT for the disabled has been steadily increasing and, in 2018 (1,171,424 requests), was about 16% higher than that in 2014 (1,011,139 requests). The share of requests not scheduled has also increased since 2014 (21.3% in 2018). The United States and South Korea have similar problems, and other countries might have similar problems in the operation of a DRT service for the disabled. Therefore, the steadily increasing demand and high operating costs for DRT services for the disabled increase the need for methods to improve the efficiency of DRT services (*6*).

This study considered that some demand for DRT for the disabled might be dispersed and others might be spatiotemporally adjacent in large-scale urban areas. If a fixed route for these demands is scheduled by finding spatiotemporally adjacent demands, a multi-capacity vehicle operating on a fixed route may be effective enough to DRT service for the disabled in large-scale urban areas. There have been studies in the DRT route design in the past; however, applying them to large-scale urban networks has encountered size limitations. This study attempts to propose a method to process data of large-scale DRT services.

LITERATURE REVIEW

In this study, a dial-a-ride problem (DARP), which is a problem for transportation service providers for the disabled and the elderly, corresponds to a vehicle routing problem with time window (VRPTW) in terms of a network problem (*7*). DARP is a variant of vehicle routing problems and NP-hard combinatorial problems in terms of computational complexity (*8*). DARP's solution methods include exact methods using mathematical models and heuristic methods to solve more realistic size problems.

Exact methods based on the concept of the branch-and-bound concept were mainly developed (*9-13*). Psarafits (1980) solved single-vehicle DARP for nine requests by formulating the problem into a dynamic program minimizing route completion time and customer dissatisfaction (*9*). Gschwind et al. (2014) solved DARP for 96 requests with branch-and-price-and-cut algorithms by considering intra-route synchronization (*12*). As exact methods cannot resolve issues regarding a large-scale dial-a-ride system,

heuristic methods that can solve significant size problems have been developed and studied to overcome the limitation of exact methods.

Heuristic methods may not find solutions with quality as high as that of exact methods; however, they do have a significant advantage over the size problem. Fu (2002) studied paratransit service systems using a parallel insertion algorithm (*14*). A first-come-first-serve assumption was used to extend the conventional heuristic algorithm with a small increase in computational time, solving 2800 requests. Xiang et al. (2006) proposed a heuristic algorithm that improves the initial solution with a local search strategy and diversification strategy and moves on with the intensification strategy (*15*). An insertion algorithm was used in the local search strategy, and the efficiency of the proposed algorithm was proved by solving a problem with 2000 requests within 10 hours using a personal computer.

Users of DRT for the disabled are classified into two types: users who need the support of a wheelchair or other mobility aid (wheelchair users), and users without any mobility aid (ambulatory users). Accordingly, the DRT service for the disabled can serve users by dividing vehicles into vehicles capable of carrying a wheelchair user and vehicles capable of carrying only an ambulatory user. Wheelchair users cannot board a vehicle serving only an ambulatory user with no wheelchair lift. DARP, which considers characteristics of DRT service, has been studied as heterogeneous DARP (HDARP) (*16-18*). Parragh et al. (2011) modified a variable neighborhood search (VNS), a heuristic method initially designed for DARP, to solve HDARP considering various types of users and vehicles in DRT service for the disabled (*16*). This study optimally solved 40 requests. Detti et al. (2016) studied HDARP in a real-world healthcare center (*17*). This study further considered heterogeneous vehicles and vehicle–patient compatibility constraints and proposed an algorithm using VNS and tabu search algorithms (TS). This study solved the problem by sampling up to 100 requests and 75 vehicles in real data. Telez et al. (2018) proposed an algorithm using ideas of a large neighborhood search (LNS) and set covering problem (SCP) to solve HDARP (*18*). The efficiency of the proposed algorithm was tested in a computational experiment, solving 295 requests extracted from the real data of users.

Studies using DARP on large-scale real DRT systems have proposed heuristic methods such as parallel insertion, tabu search, VNS, and LNS. Studies considering the properties of the DRT service for the disabled have solved user requests sampled from real data. This study proposes a cluster-based fixed route for a multi-capacity vehicle based on past usage data to improve the efficiency of the real DRT service for the disabled in large-scale urban areas and evaluates the efficiency of a cluster-based fixed route in both toy and large-scale networks.

METHODOLOGY

Problem Description

This study aims to design a fixed route of a multi-capacity vehicle based on a cluster technique by using historical data in HDARP. Components of DRT service for the disabled are users, vehicles, operation center, and network.

There are two types of DRT service for the disabled (u_i): wheelchair users and ambulatory users, depending on the service requirements, with respect to the physical characteristics of users. Wheelchair users must be serviced by a vehicle equipped with a wheelchair lift. As ambulatory users can ride any vehicle, all vehicles are available to ambulatory users. The requests by users received at the operation center include pick-up location (origin), drop-off location (destination), desired leaving (pick-up) time, and user type. The DRT route should be designed according to the time schedule that can pick up the user after the desired leaving time because a time window constraint is caused by the desired leaving time. The request of individual users is the element of the set of requests (**R**), and the notation used in this study is as follows:

$$\mathbf{R} = \{r_1, r_2, \dots, r_{N_r}\}, r_i = (o_i, d_i, e_i, u_i)$$

where N_r is the number of requests, o_i is the origin of the *i*-th request (user), $i \in \boldsymbol{N} = \{1, \dots, N_r\}$, d_i is the destination of the *i*-th request (user), e_i is the earliest (desired leaving) time of the *i*-th user, and u_i is the type of the *i*-th user ($u_i = 0$: an ambulatory user, $u_i = 1$: a wheelchair user).

Vehicles of a DRT service for the disabled are divided into a single-capacity vehicle and a multi-capacity vehicle. A single-capacity vehicle consists of a vehicle equipped with a wheelchair lift capable of serving wheelchair users and a vehicle without a wheelchair lift. Depending on the capacity and the seat configuration of a multi-capacity vehicle, the number of wheelchair users and ambulatory users who can board a multi-capacity vehicle are determined. The operation center dispatches vehicles for DRT service for the disabled according to service requirements (origin, destination, desired leaving time, and user type) of all users on the FCFS.

Components of the network for DRT service consist of stations (**S**) and travel time between stations (**TT**). The station represents the spatial position of the pick-up and drop-off locations. The notation of stations in this study is as follows: $\mathbf{S} = \{s_1, s_2, \dots, s_{N_s}\}$, where N_s is the number of stations. The notation of travel time (**TT**) between stations of this study is as follows:

$$\mathbf{TT} = \begin{pmatrix} tt_{11} & \cdots & tt_{1N_s} \\ \vdots & \ddots & \vdots \\ tt_{N_s1} & \cdots & tt_{N_sN_s} \end{pmatrix}$$

This study consists of four steps (**Figure 1**):

Step 1. Constructing initial routes: This step constructs initial routes in which a vehicle can move users by using the set of requests and matrix of travel time.

Step 2. Measuring similarity: This step computes the similarity of each position in the two routes of the initial routes and then calculates the similarity between the two routes.

Step 3. Clustering routes: This step clusters routes that are close to each other based on the matrix of distance for routes by using density-based spatial clustering of applications with noise (DBSCAN).

Step 4. Designing cluster-based route: This step creates a fixed route that can move many users at the same time by using routes in the clusters obtained in Step 3 as input.

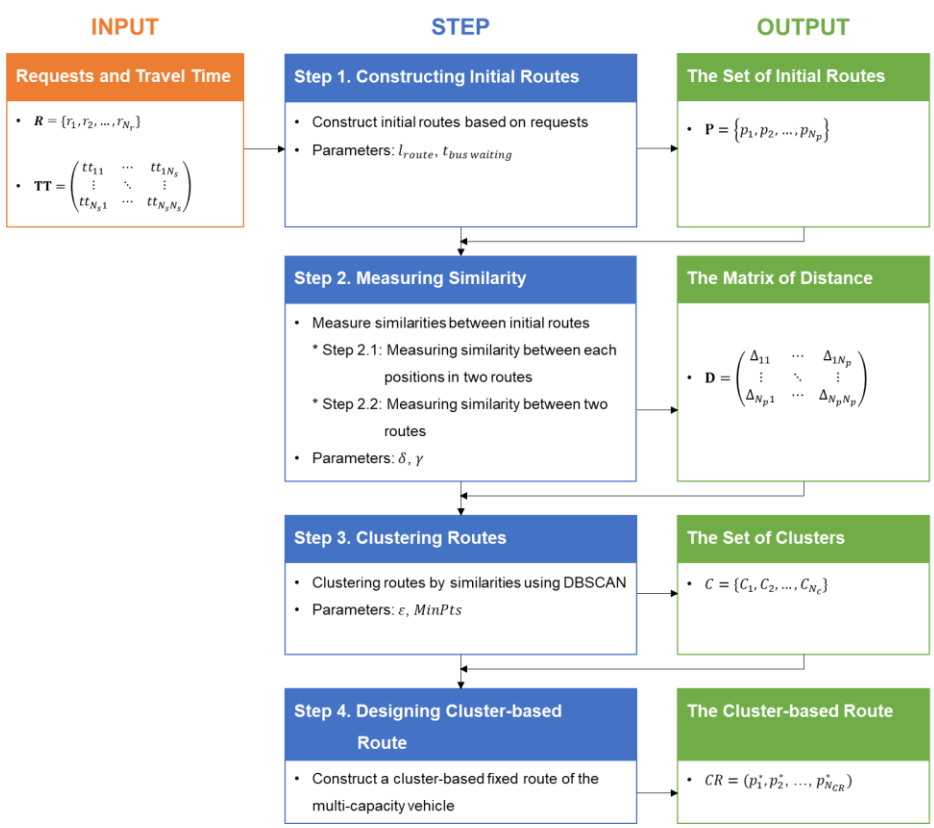

FIGURE 1 Framework of Cluster-Based Route Design

Step 1. Constructing Initial Routes

Constructing initial routes is an iterative process that creates a route where vehicles could be operated continuously from historical requests consisting of desired leaving times, pick-up stations, and drop-off stations (O-D pairs). The input of Step 1 is the set of requests (**R**) and the matrix of travel time (**TT**). Step 1 has two parameters: the maximum number of requests that can be served (l_{route}); the maximum allowable waiting time interval of vehicle for pick-up ($t_{veh\ waiting}$).

The first process of Step 1 constructs a route for all requests sequentially based on the desired leaving time if there is a request within $t_{veh\ waiting}$ at the destination of the request. This process creates another initial route if the number of requests served by the route is greater than l_{route}. The process of constructing initial routes is terminated when no more initial routes can be created. The algorithm of constructing initial routes is as follows:

Algorithm 1. Scheme of Procedure for Constructing Initial Routes

Input : The set of requests (**R**), The matrix of travel time (**TT**)
Output : The set of routes (**P**)
Parameter : l_{route}, $t_{veh\ waiting}$
Procedure :

```
Set Route_ID l = 1;
Mark all requests in R as not visited;
For each request r_i in R,
    If r_i is not visited,
        Initialize p_l (p_l = ∅);
        Set Point_ID n = 1;
        Insert the origin and desired leaving time of r_i into route p_l
        Set k = i;
        While all requests are not searched, do;
            For each request r_j in R,
                If r_j is not visited and e_k + tt_{o_k d_j} ≤ e_j ≤ e_k + tt_{o_k d_j} + t_{veh waiting},
                    Mark r_k and r_j as visited;
                    Insert the origin and desired leaving time of r_j into route p_l;
                    Set k = j, j = 1;
                    Break;
                If n(p_l) = l_route,
                    Break;
            If n(p_l) > 1,
                Insert the destination and desired leaving time of r_j into route p_l;
                Set Route_ID l = l + 1;
Return P;
```

The output of route generation is the set of routes (P) as follows:

$$\mathbf{P} = \left\{p_1, p_2, \dots, p_{N_p}\right\},\ p_i = \left(p_1^i, p_2^i, \dots, p_{N_{r_i}}^i\right), p_k^i = \left(s_k^i, t_k^i\right)$$

where N_P is the number of routes, p_k^i is the *k*-th position of the *i*-th route, s_k^i is the station in the *k*-th postion of the *i*-th route, t_k^i is the departure time on the *k*-th postion of the *i*-th route.

Step 2. Measuring Similarity

Measuring similarity consists of two substeps: measuring similarity between each position in two routes; measuring similarity between two routes. This study modified some parts of an existing study (*19*) using the similarity measurement to identify traffic patterns using trajectories according to the DRT route.

Step 2-1: Measuring Similarity between Each Position in Two Routes

This study uses a similarity measure based on the longest common subsequence (LCS), which can measure the gap between sequences because generated routes are sequences composed of station and departure time. The LCS finds the LCS of two routes and matches the positions of those routes. In Step 2-1, the inputs are the difference in departure time of each position ($\left|t_k^i - t_l^j\right|$) on the two routes and the travel time ($tt_{s_k^i s_l^j}$) between each position in two routes. The parameter (δ) in Step 2-1 is the maximum allowable travel time that can be judged by similar routes. The difference in departure time between two positions is a measure of the similarity on the same station; further, the travel time ($tt_{s_k^i s_l^j}$) between each position in the two routes measures the similarity on different stations. In order to match positions with high similarity, this study compares the larger of the two values indicating the spatiotemporal distance with the value of δ.

It assumes that there are two routes of $p_i = (p_1^i, p_2^i, \dots, p_N^i)$ and $p_j = (p_1^j, p_2^j, \dots, p_M^j)$ with the lengths of $N\ (= N_{r_i})$ and $M\ (= N_j)$. The p_k^i and p_l^j on the two routes are matched if the following conditions are satisfied:

$$\max\left(\left|t_k^i - t_l^j\right|, tt_{s_k^i s_l^j}\right) < \delta. \tag{1}$$

The similarity of two matched positions p_k^i and p_l^j are calculated as follows:

$$simPos\left(p_k^i, p_l^j\right) = \begin{cases} 0 & \text{if } \max\left(\left|t_k^i - t_l^j\right|, tt_{s_k^i s_l^j}\right) > \delta \\ 1 - \frac{1}{\delta} \times \max\left(\left|t_k^i - t_l^j\right|, tt_{s_k^i s_l^j}\right) & \text{otherwise.} \end{cases} \tag{2}$$

The purpose of the LCS is to obtain the LCS sequence composed of points with high similarity among the matched points. This study uses the following recursive function to obtain the LCS sequence using a dynamic programming model proposed by a previous study (*19*):

$$p_{i(n)} = (p_1^i, p_2^i, \dots, p_n^i),\ p_{j(m)} = (p_1^j, p_2^j, \dots, p_m^j) \text{ where } n \le N \text{ and } m \le M$$

$$LCS\left(p_{i(n)}, p_{j(m)}\right) = \begin{cases} 0 & \text{if } n = 0 \text{ or } m = 0 \\ \max\begin{pmatrix} LCS\left(p_{i(n)}, p_{j(m)}\right) + simPos\left(p_k^i, p_l^j\right), \\ LCS\left(p_{i(n)}, p_{j(m-1)}\right), \\ LCS\left(p_{i(n-1)}, p_{j(m)}\right) \end{pmatrix} & \text{otherwise.} \end{cases} \tag{3}$$

The function $LCS\left(p_{i(n)}, p_{j(m)}\right)$ derives the LCS sequence with the highest similarity up to the *n*-th position of the *i*-th route and the *m*-th position of the *j*-th route. The output of similarity measurement between each position in two routes is the set of LCS (**L**) in the routes, and the notation of the set of LCS is as follows: $L_{ij} = \{q_1^{ij}, \dots, q_{N_{ij}}^{ij}\}$ where, N_{ij} is the number of matched points.

Step 2-2: Measuring Similarity Between Two Routes

Similarity between two routes is calculated using the set of LCS (**L**) as input data in Step 2-2. The *m*-th points matched by the function $LCS\left(p_{i(N)}, p_{j(M)}\right)$ are represented by q_m^{ij} and are in the *i*-th route and the *j*-th route, respectively. If p_k^i and p_l^j on two routes are matched, then q_m^{ij} is as follows: $q_m^{ij} = \left(q_m^i, q_m^j\right) = \left(p_{(m)}^i, p_{(m)}^j\right) = \left(p_k^i, p_l^j\right)$, where q_m^i is the *m*-th matched point in the *i*-th route, and q_m^j is the *m*-th matched point in the *j*-th route. The LCS (L_{ij}), composed of matched points of two routes, is a subset

of each route (i, j). The length of sequences (L_{ij}^{i}, L_{ij}^{j}) of matched points in each route is defined as the sum of travel times between stations of each point as follows:

$$lenSeq\left(L_{ij}^{i}\right) = \sum_{l} tt_{s_{(l)}^{i} s_{(l+1)}^{i}} \text{ and } lenSeq\left(L_{ij}^{j}\right) = \sum_{l} tt_{s_{(l)}^{j} s_{(l+1)}^{j}}.$$

The parameters of similarity measurement between two routes, γ, means the minimum allowable length for LCS is valid. Even if the similarity between routes is large, the concept of a minimum allowable length should be introduced if the LCS is not long enough to be considered a cluster-based route. In a given LCS, similarity between two routes (p_i, p_j) is calculated as follows:

$$simRoute\left(p_i, p_j\right) = \begin{cases} 0 & \text{if } \min\left(lenSeq\left(L_{ij}^{i}\right), lenSeq\left(L_{ij}^{j}\right)\right) < \gamma \\ \frac{lenSeq\left(L_{ij}^{i}\right)}{\min\left(lenSeq(p_i), lenSeq(p_i)\right)} & \text{if } lenSeq\left(L_{ij}^{i}\right) < lenSeq\left(L_{ij}^{j}\right) \\ \frac{lenSeq\left(L_{ij}^{j}\right)}{\min\left(lenSeq(p_i), lenSeq(p_j)\right)} & \text{otherwise.} \end{cases} \quad (4)$$

Finally, using the similarity between two routes (p_i, p_j) defined above, we can compute the distance between two routes (p_i, p_j) used as input for clustering. The distance between two routes (p_i, p_j) is a normalized value between 0 and 1, as follows:

$$distRoute\left(p_i, p_j\right) = \Delta_{ij} = 1 - simRoute\left(p_i, p_j\right). \quad (5)$$

The distances between all routes calculated through Step 2 are expressed in matrix form as follows:

$$\mathbf{D} = \begin{pmatrix} \Delta_{11} & \cdots & \Delta_{1N_p} \\ \vdots & \ddots & \vdots \\ \Delta_{N_p 1} & \cdots & \Delta_{N_p N_p} \end{pmatrix}.$$

Step 3. Clustering Routes

In this study, density-based spatial clustering of applications with noise (DBSCAN) is used as a clustering algorithm (*20*). DBSCAN is an algorithm that does not require setting the number of clusters in advance; further, it effectively prevents noise data from being affected by clustering and shows excellent performance in large-scale applications (*21*). In the DBSCAN algorithm, the distance matrix (**D**) calculated by the similarity measure between routes in Step 2 is used for input. The algorithm of DBSCAN has two parameters: ϵ, *MinPts*. In this study, ϵ is defined as the maximum route-to-route distance allowed for a similar route. *MinPts* is related to the capacity of a multi-capacity vehicle, as it is the minimum number of similar routes required for a cluster-based route. Key definitions and notation of the DBSCAN algorithm used in this study are modified in (21) and are as follows:

Definition 1 (density of a route) The density of a route p_i is the number of routes in area which set p_i as the center route and ϵ as the radius.

Definition 2 (ϵ-neighborhood of a route) The ϵ-neighborhood of a route p_i is defined as $N_\epsilon(p_i) = \{p_j \in \mathbf{P} \mid distRoute\left(p_i, p_j\right) \leq \epsilon\}$.

Definition 3 (directly density-reachable) A route p_j is said to be directly density-reachable from a route p_i with respect ϵ and *MinPts* if $p_j \in N_\epsilon(p_i)$ and $|N_\epsilon(p_i)| \geq$*MinPts* (cluster-based route condition)

Definition 4 (density-reachable) A route p_j is said to be density-reachable from a route p_i with respect ϵ and *MinPts* if there is a chain of routes $p_1, \ldots, p_k, p_{k+1}, \ldots, p_{N_p}$, $p_1 = p_j$, $p_{N_p} = p_i$ such that p_k is directly density-reachable from p_{k+1}.

Definition 5 (density-connected) A route p_j is said to be density-connected to a route p_i with respect ϵ and *MinPts* if there is a route p_k such that both p_i and p_j are density-reachable from p_k with respect to ϵ and *MinPts*.

Definition 6 (cluster) A cluster $\boldsymbol{C}_n$ with respect to ϵ and *MinPts* is a nonempty subset of **P** satisfying the following conditions:

1) $\forall p_i, p_j \in \mathbf{P}$, if $p_i \in \boldsymbol{C}_n$ and p_j is density-reachable from p_i with respect to ϵ and *MinPts*, then $p_j \in \boldsymbol{C}_n$ (maximality)

2) $\forall p_i, p_j \in \boldsymbol{C}_n, p_i$ is density-connected to p_j with respect to ϵ and *MinPts*, then $p_j \in \boldsymbol{C}_n$ (Connectivity)

Definition 7 (noise) Noise is the set of routes **P** not belonging to any cluster $\boldsymbol{C}_n$ (n = 1, …, N_c)

The DBSCAN algorithm used in this study is as follows.

Algorithm 2. Scheme of DBSCAN Algorithm

```
Input      : The set of routes (P), The matrix of distance (D)
Output     : The set of clusters (C)
Parameter  : ϵ, MinPts
Procedure  :
DBSCAN(P, ϵ, MinPts)
    Set Cluster_ID n = 1;
    Mark all routes in P as unclassified;
    For each route p_i in P,
        If p_i is unclassified,
            Mark p_i as classified;
            Get N_ϵ(p_i) using D;
            Call ExpandCluster(P, p_i, ClusterID, Q, D, ϵ, MinPts);
            Set Cluster_ID n = n + 1;
    Return C;
ExpandCluster(P, p_i, Cluster_ID, Q, D, ϵ, MinPts)
    If |N_ϵ(p_i)| < MinPts,
        Mark p_i as noise;
    Else,
        Assign i to all routes in N_ϵ(p_i);
        Insert N_ϵ(p_i) into queue Q;
        While queue Q is not empty, do;
            Remove p_i from queue Q;
            For each route p_j in N_ϵ(p_i),
                If p_j is unclassified,
                    If p_j is noise,
                        Assign Cluster_ID to p_j;
                    Else,
                        Insert p_j into queue Q;
```

The output of Step 3 is the set of clusters (**C**) in which routes with high spatiotemporal similarity are assembled and as follows:

$$\mathbf{C} = \{C_1, C_2, \dots, C_{N_c}\},\ C_m = \{\tilde{p}_1^{C_m}, \tilde{p}_2^{C_m}, \dots, \tilde{p}_{|C_m|}^{C_m}\}$$

where N_c is the number of clusters, $\tilde{p}_i^{C_m}$ is the i-th route in the m-th cluster C_m, $|C_m|$ is the number of routes in the m-th cluster C_m.

Step 4. Designing Cluster-based Route

This study designs the cluster-based route by connecting requests with frequent O-D pairs in the same time zone based on all requests belonging to clusters extracted by the DBSCAN algorithm. This study

uses requests for all clustered routes in Step 3 as inputs. The procedure for designing the cluster-based route consists of the following steps.

Algorithm 3. Scheme of procedure for designing the cluster-based route

1) Separate requests by time: All requests in clusters are separated by time period ($\mathrm{H} = \{h | h = 0, 1, \ldots, 23\}$) to form a separate set (R_h).
2) Count OD pairs: After the same O-D pairs are counted for each time period, the desired leaving time of O-D pair is acquired using the average value of the desired leaving times of the same O-D pairs.
3) Construct cluster-based route: The O-D pairs are included in the cluster-based route in descending order of frequency until multi-capacity vehicle can move in that time period (h). This process is repeated in the next time period ($h + 1$).

The output of Step 4 is the vehicle dispatch plan of the cluster-based route as follows:

$$\boldsymbol{CR} = \{p_1^*, p_2^*, \ldots, p_{N_{CR}}^*\}, \qquad p_k^* = (s_k^*, t_k^*)$$

where N_{CR} is the length of a cluster-based route, p_k^* is the k-th position of a cluster-based route, s_k^* is the station in the k-th position of cluster-based route, t_k^* is the departure time on the k-th position of cluster-based route.

CASE STUDY

The purpose of the case study is to verify the efficiency of a cluster-based route in both toy and large-scale networks. This section includes performance evaluation, analysis in a toy network, and analysis in a large-scale network. Performance evaluation introduces performance measures, evaluation methods, and operation strategies.

Performance Evaluation

Performance measures are indicators to evaluate the performance of the simulation. Mean and standard deviations of user waiting time were used in this study. Analytical method and simulation are used to evaluate the performance of the practical operation strategy (*22*). The performance of Strategy 1 and Strategy 2 is evaluated with simulation, and the performance of Strategy 3 is evaluated with mathematical formulation. Agent-based simulation of DRT service was implemented in the JAVA programming environment. Operation strategies are the rules for dispatching vehicles to users.

Three operation strategies were evaluated in this study.

1) Strategy 1 (Current): FCFS with single capacity vehicles
   (priority of dispatch: the nearest single-capacity vehicle in idle)
2) Strategy 2 (Cluster-based): FCFS with single-capacity vehicles and a multi-capacity vehicle
   (priority of dispatch: first, a multi-capacity vehicle if the origin node of user is equal to one of nodes in a cluster-based route;
   second, the nearest single-capacity vehicle in idle)
3) Strategy 3 (Exact solution): Dispatching of Strategy 1 with hindsight

Finding the exact solution of the current operation in a large-scale network is impossible due to the size problem. For this reason, the performance of Strategy 3 is evaluated in a toy network but not in a large-scale real network.

This study formulated a mathematical model to implement Strategy 3. The notation and mathematical model used in this study were extended from the basic mathematical model of the DARP introduced by (*8*). The DARP problem with n users (or requests) to be served is modeled on a graph $G = (V, A)$. Set V of vertices contains set $P = \{1, \ldots, n\}$ of pick-up vertices, set $D = \{n + 1, \ldots, 2n\}$ of drop-off vertices, and origin and destination depots (0 and $2n + 1$, respectively). The set of arcs is denoted by A.

Set of types of users (R) and set of types of vehicles (M) are defined to represent the hererogeneous fleet and requests. The user i is characterized by origin vertex i and destination vertex $n+i$. The type of the user is denoted by $r_i \in R$ ($r_i = 0$: ambulatory user, $r_i = 1$: wheelchair user). Set of vehicles and vehicle type is denoted by K and $m_i \in M$ ($m_k = 0$: single-capacity vehicle without a wheelchair lift, $m_k = 1$: single-capacity vehicle with a wheelchair lift), respectively. Vehicle k is characterized by capacity w_k and maximum total duration T_k. Each vertex i has load q_i, service duration d_i, and the time window $[e_i, l_i]$ (the earliest and latest time, respectively, at which service may begin at vertex i). Each arc (i, j) is associated with routing cost c_{ij} and travel time t_{ij}. Cost of time is denoted by c_{time}. In addition, maximum ride time of a user L, large number B, and weight α is defined for the mathematical formulation.

The following variables are introduced for the mathematical formulation:

x_{ij}^k is a binary variable which is equal to 1 if vehicle k travels from vertex i to vertex j.

u_i^k is the time at which vehicle k begins service at vertex i.

w_i^k is the load of vehicle k upon leaving vertex i.

L_i^k is the ride time of user i on vehicle k.

Mathematical formulation of Strategy 3 is represented by mixed-integer linear programming (MILP) as follows:

$$\min \alpha \sum_{k\in K}\sum_{i\in V}\sum_{j\in V} c_{ij}^k x_{ij}^k + (1-\alpha)\sum_{k\in K}\sum_{i\in P} c_{time}(u_i^k - e_i) \tag{6}$$

s.t.

$$\sum_{k\in K}\sum_{j\in V} x_{ij}^k = 1 \qquad \forall\, i \in P \tag{7}$$

$$\sum_{j\in V} x_{ij}^k - \sum_{j\in V} x_{n+ij}^k = 0 \qquad \forall\, i \in P, k \in K \tag{8}$$

$$\sum_{j\in V} x_{0j}^k = 1 \qquad \forall\, k \in K \tag{9}$$

$$\sum_{j\in V} x_{ji}^k - \sum_{j\in V} x_{ij}^k = 0 \qquad \forall\, i \in P \cup D, k \in K \tag{10}$$

$$\sum_{j\in V} x_{i,2n+1}^k = 1 \qquad \forall\, k \in K \tag{11}$$

$$u_j^k \geq \left(u_i^k + d_i + t_{ij}\right) x_{ij}^k \qquad \forall\, i \in V, j \in V, k \in K \tag{12}$$

$$w_j^k \geq \left(w_i^k + q_i\right) x_{ij}^k \qquad \forall\, i \in P, k \in K \tag{13}$$

$$L_i^k = u_{n+i}^k - (u_i^k + d_i) \qquad \forall\, i \in P, k \in K \tag{14}$$

$$u_{2n+1}^k - u_0^k < T_k \qquad \forall\, k \in K \tag{15}$$

$$e_i \leq u_i^k \leq l_i \qquad \forall\, i \in V, k \in K \tag{16}$$

$$t_{i,n+i} \leq L_i^k \leq L \qquad \forall\, i \in P, k \in K \tag{17}$$

$$max\,\{0, q_i\} \leq w_i^k \qquad \forall\, i \in V, k \in K \tag{18}$$

$$w_i^k \leq min\,\{w_k, w_k + q_i\} \qquad \forall\, i \in V, k \in K \tag{19}$$

$$x_{ij}^k (m_k - r_i) \geq 0 \qquad \forall\, i \in V, j \in V, k \in K, r_i \in R, m_k \in M \tag{20}$$

$$x_{ij}^k \in \{0, 1\} \qquad \forall\, i \in V, j \in V, k \in K \tag{21}$$

The objective function (6) is to minimize the weighted sum of users' waiting time and vehicles' travel cost. Constraints (7) and (8) request to be served exactly once by the same vehicle. Constraints (9), (10), and (11) state that the route of each vehicle k starts at the origin depot and ends at the destination depot. Constraints (12) and (13) ensure that time and load variables are consistent. Constraint (14) defines the ride time of each user, and constraint (15) limits the duration of each routes. Constraint (16) is the time window constraint, and constraint (17) limits the ride time of each trip. Constraints (18) and (19) are the capacity constraints. Constraint (20) limits wheelchair users to be serviced by special vehicles only. Constraints (21) define the decision variable x_{ij}^k. Nonlinear constraints (12) and (13) are transformed into linear constraints with large number B as follows: $u_j^k \geq u_i^k + d_i + t_{ij} - B(1 - x_{ij}^k)$, $w_j^k \geq w_i^k + q_i - B(1 - x_{ij}^k)$. The mathematical model was coded into a JAVA programming environment and solved by IBM ILOG CPLEX version 12.7.

Analysis in Toy Network

The performances of Strategy 1, Strategy 2, and Strategy 3 were evaluated and compared in this research. The toy network consists of one starting depot (1st node), one returning depot (12th node), and 10 station nodes (2nd, …, 11th node). The operation hour of the operation center in the toy network is 10 hours.

The size of the fleets used in the toy network is five, and the types of those consist of as follows: a single-capacity vehicle equipped with a wheelchair lift, a single-capacity vehicle not equipped with a wheelchair lift, and a multi-capacity vehicle (four wheelchair users and two ambulatory users). Five vehicles operated in Strategy 1 and Strategy 3 consist of three single-capacity vehicles equipped with a wheelchair lift and two single-capacity vehicles not equipped with a wheelchair lift. Five vehicles operated in Strategy 2 consist of two single-capacity vehicles equipped with a wheelchair lift, two single-capacity vehicles not equipped with a wheelchair lift, and a multi-capacity vehicle.

The number of users in the toy network is 40, with 12 ambulatory users (user type = 0) and 28 wheelchair users (user type = 1). Requests of the toy network users are summarized in **Table 1**.

TABLE 1 Detail Requests in the Toy Network

No.	O-D	User type	Pickup time window	No.	O-D	User type	Pickup time window
1	4-3	1	09:40-10:10	21	6-5	1	13:05-13:35
2	11-10	1	10:20-10:50	22	7-6	1	12:05-12:35
3	2-3	1	09:35-10:05	23	4-8	0	13:20-13:50
4	3-11	1	10:15-10:45	24	7-6	0	12:30-13:00
5	10-8	1	11:00-11:30	25	7-5	1	13:10-13:40
6	3-11	1	10:10-10:40	26	4-3	1	13:15-13:45
7	4-11	0	09:50-10:20	27	3-8	1	13:55-14:25
8	2-11	1	09:45-10:15	28	7-5	1	12:10-12:40
9	10-8	0	10:55-11:25	29	11-9	1	14:25-14:55
10	11-10	0	10:50-11:20	30	9-10	1	15:05-15:35
11	11-10	1	10:25-10:55	31	8-7	0	12:35-13:05
12	9-8	1	11:20-11:50	32	5-4	1	13:40-14:10
13	11-9	0	10:45-11:15	33	3-11	1	13:50-14:20
14	8-7	1	11:55-12:25	34	8-9	1	14:30-15:00
15	9-8	1	12:00-12:30	35	5-3	1	13:45-14:15
16	10-9	1	11:25-11:55	36	11-8	0	14:55-15:25
17	5-4	1	12:45-13:15	37	4-3	1	14:15-14:45
18	6-4	0	12:40-13:10	38	3-11	1	14:20-14:50
19	8-7	0	11:35-12:05	39	10-8	0	15:40-16:10
20	8-7	1	11:30-12:00	40	3-11	0	14:50-15:20

The following parameters were defined based on the DRT service the disabled in Seoul and used in the toy network: l_{route} = 10 requests, $t_{veh\ waiting}$ = 15 minutes, δ = 30 minutes, γ = 60 minutes, ϵ = 0.1, *MinPts* = 3 points, T_k = 10 hours ($\forall\ k \in K$), Q_k= 1 person ($\forall\ k \in K$), c_{time} = \$10 per hour, L = 1 hour, B = 1000, α = 0.5. The travel time matrix of the depots and nodes in the toy network is as follows:

$$\mathbf{TT} = \begin{pmatrix} 0 & 20 & 20 & 20 & 20 & 30 & 50 & 60 & 60 & 50 & 40 & 80 \\ 20 & 0 & 20 & 30 & 40 & 50 & 50 & 50 & 50 & 40 & 30 & 60 \\ 20 & 20 & 0 & 20 & 20 & 30 & 40 & 40 & 40 & 30 & 20 & 60 \\ 20 & 30 & 20 & 0 & 20 & 30 & 40 & 40 & 40 & 40 & 30 & 60 \\ 20 & 40 & 20 & 20 & 0 & 20 & 30 & 30 & 40 & 40 & 30 & 60 \\ 30 & 50 & 30 & 30 & 20 & 0 & 20 & 30 & 30 & 40 & 30 & 60 \\ 50 & 50 & 40 & 40 & 30 & 20 & 0 & 25 & 30 & 40 & 30 & 40 \\ 60 & 50 & 40 & 40 & 30 & 30 & 25 & 0 & 20 & 30 & 30 & 20 \\ 60 & 50 & 40 & 40 & 40 & 30 & 30 & 20 & 0 & 20 & 20 & 20 \\ 50 & 40 & 30 & 40 & 40 & 40 & 40 & 30 & 20 & 0 & 20 & 30 \\ 40 & 30 & 20 & 30 & 30 & 30 & 30 & 30 & 20 & 20 & 0 & 35 \\ 80 & 60 & 60 & 60 & 60 & 60 & 40 & 20 & 20 & 30 & 35 & 0 \end{pmatrix}.$$

In Step 1, five routes were generated based on 40 requests, and the similarity measure between routes was calculated in Step 2. In Step 3, a cluster was derived from five routes, and three routes were included in a cluster. In Step 4, a cluster-based route touring eight stations from 09:00 to 19:00 was derived from 36 requests belonging to three routes by applying the cluster-based route design algorithm [**Figure 2 (a)**].

Figure 2 shows the performance results of strategies in a toy network. In terms of average waiting time for all users, the average waiting time for all users in Strategy 2 decreased to 63.67% of that in the current operation. The average waiting time for all users in Strategy 3 decreased to 12.17% of that in the current operation. For users using a cluster-based route, the average waiting time of users on a cluster-based route in Strategy 2 decreased to 18.79% of that in the current operation. The average waiting time of users on a cluster-based route in Strategy 3 decreased to 18.79% of that in the current operation. The standard deviation of users' waiting time in Strategy 3 was the smallest, and that in Strategy 1 was the largest. This means that the standard deviation of users' waiting time decreases as the service is operated more efficiently.

Figure 2 (c) shows the cumulative waiting time of all users as the number of users increases. As the number of users increases, the difference in cumulative waiting time among all users of the three strategy performances becomes larger. **Figure 2 (d)** shows the waiting time of users on a cluster-based route. In **Figure 2 (d)**, some users of Strategy 1 waited for more than two hours, but these users were found to be less than 30 minutes in Strategy 2 and Strategy 3.

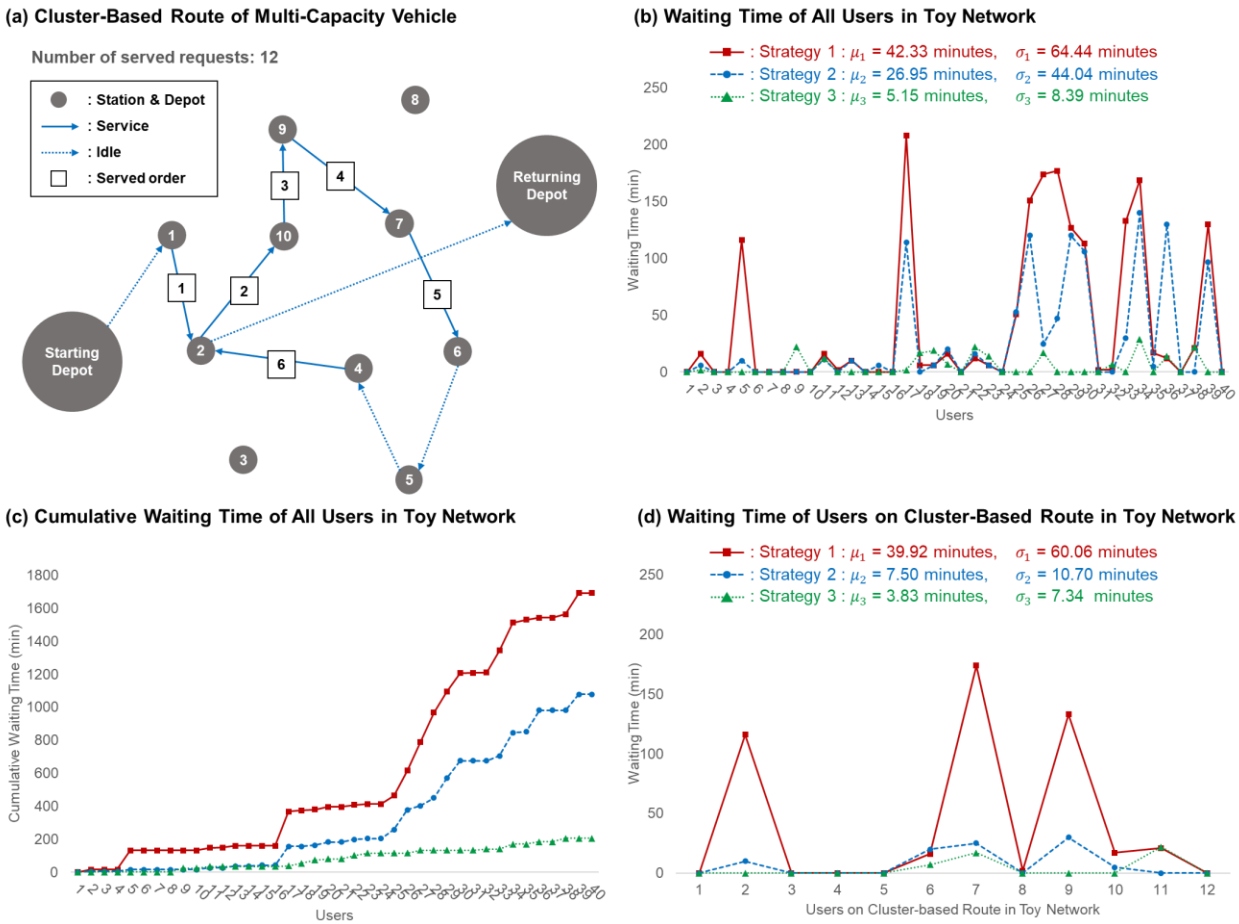

FIGURE 2 A Cluster-Based Route and Results of Operation Strategies in the Toy Network

Analysis in a Large-Scale Network: DRT for the Disabled in Seoul

The DRT service for the disabled in Seoul, which started in 2013, currently operates 487 vehicles: 436 single-capacity vehicles with a wheelchair lift, 50 single-capacity vehicles without a wheelchair lift, and a multi-capacity vehicle. A multi-capacity vehicle is designed to accommodate both four wheelchair users and two ambulatory users. Vehicles operating in DRT service for the disabled in Seoul are shown in **Figure 3**. The operation center supplies service in the order of desired leaving time and reception time for 24 hours on FCFS and dispatches the vehicle at a close distance.

Source : https://www.sisul.or.kr/open_content/calltaxi/
http://www.seouland.com/arti/society/society_general/55.html, http://heraldk.com

FIGURE 3 Vehicles Operated in DRT Service for the Disabled in Seoul

This study collected actual data from 1,251,271 trips (January 1, 2016 ~ December 31, 2016) for the total number of users in 2016. Twenty percent of user requests were advance requests, and the average number used per day was 3,419 trips. This study calculated travel time by using all of the data for one year to improve the reality of travel time. For the performance evaluation of a cluster-based route, this study used 20 weekdays during July 2016. Ten days (day 1 ~ day 10) were used to find a cluster-based route, and the remaining 10 days (day 11 ~ day 20) were used to evaluate the performance of the identified cluster-based route. The values of components and parameters used in this study are shown in **Table 2**.

TABLE 2 Values of Components and Parameters Used in a Large-Scale Network

Components		**Value**
Request (user)		20 weekdays in July, 2016 54,098 requests (wheelchair users) 29,751 requests (ambulatory users)
Vehicle		436 vehicles (single-capacity vehicle with a wheelchair lift) 50 vehicles (single-capacity vehicle without a wheelchair lift) 1 vehicle (multi-capacity vehicle)
Operation Center		24 hours (operation hours)
Stations		716 stations
Travel Time		Travel time estimated by actual data
Parameters		**Value**
Step 1	l_{route}	10 requests
	$t_{veh\ waiting}$	15 minutes
Step 2	δ	30 minutes
	γ	60 minutes
Step 3	ϵ	0.1
	$MinPts$	6 points

In Step 1, 5,569 routes were generated based on 42,281 requests for 10 days, and the similarity measure between routes was calculated in Step 2. In Step 3, 32 clusters were derived from 5,569 routes, and 2,044 routes were included in 32 clusters. That is, 3,525 routes were treated as noise in DBSCAN. In Step 4, a cluster-based route touring 21 stations from 07:00 to 17:00 was derived from using 5,224 requests belonging to 2,044 routes by applying the cluster-based route design algorithm. The operating time of the cluster-based route was confirmed in the time period when 88.5% of the total demand occurs, and the cluster-based route was designed in a time period with high demand.

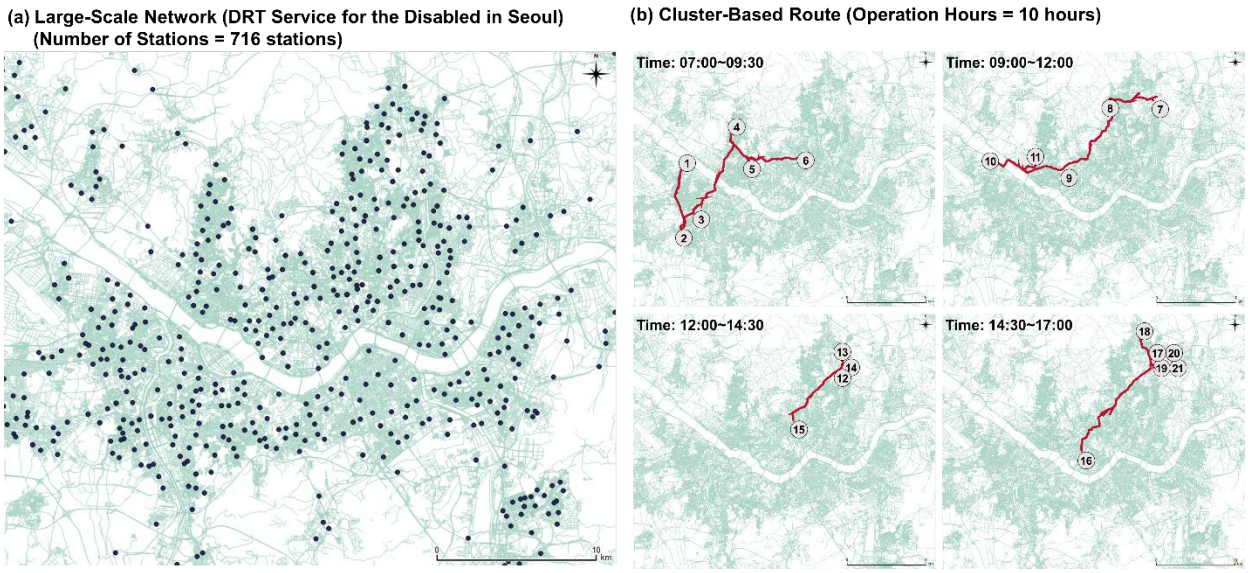

FIGURE 4 Network and Cluster-Based Route Designed in the Large-Scale Network

Strategy 1 and Strategy 2 were compared and analyzed by simulation. In order to reflect the randomness of the vehicle dispatch, each scenario was simulated 30 times. Calibration was performed to ensure reality of the network. The daily average user waiting time and average user waiting time during the operation time of the cluster-based route were compared with the actual data for the simulation calibration. The result of the calibration is as follows: root mean square error = 0.96 min; mean absolute percentage error = 1.77 %.

Strategy 1 operated 486 vehicles (436 single-capacity vehicles with a wheelchair lift and 50 single-capacity vehicles without a wheelchair lift). Strategy 2 operated 486 vehicles (435 single-capacity vehicles with a wheelchair lift, 50 single-capacity vehicles without a wheelchair lift, and a multi-capacity vehicle).

Simulation results of Strategy 1 and Strategy 2 were compared to evaluate the performance of the cluster-based route (**Figure 5**). The cluster-based fixed routes served an average of 26.65 requests from day 1 to 10 and an average of 17.77 from day 11 to day 20 [**Figure 5 (a)**]. This is because the cluster created by the DBSCAN algorithm is partially overfitted. Although the performance from day 11 to day 20 in the cluster was lower than the performance from day 1 to day 10, a multi-capacity vehicle (17.77 users) provided an average of 82.66% more users than single-capacity vehicles (9.69 users). A multi-capacity vehicle reduced user waiting time by 81.2% over single-capacity vehicles.

With the introduction of a multi-capacity vehicle, it was confirmed that the waiting time of users of the cluster-based route dramatically decreased. Improvement of the overall system efficiency was a mere 1.2%. However, considering the fact that this is the result of introducing only one multi-capacity vehicle into a fleet of 486 vehicles, the additional introduction of the multi-capacity vehicle is expected to increase the effect. Consequently, we demonstrated that the proposed methodology can design a cluster-based route that can introduce a fixed route by effectively measuring time and space similarity in dispersed historical request data.

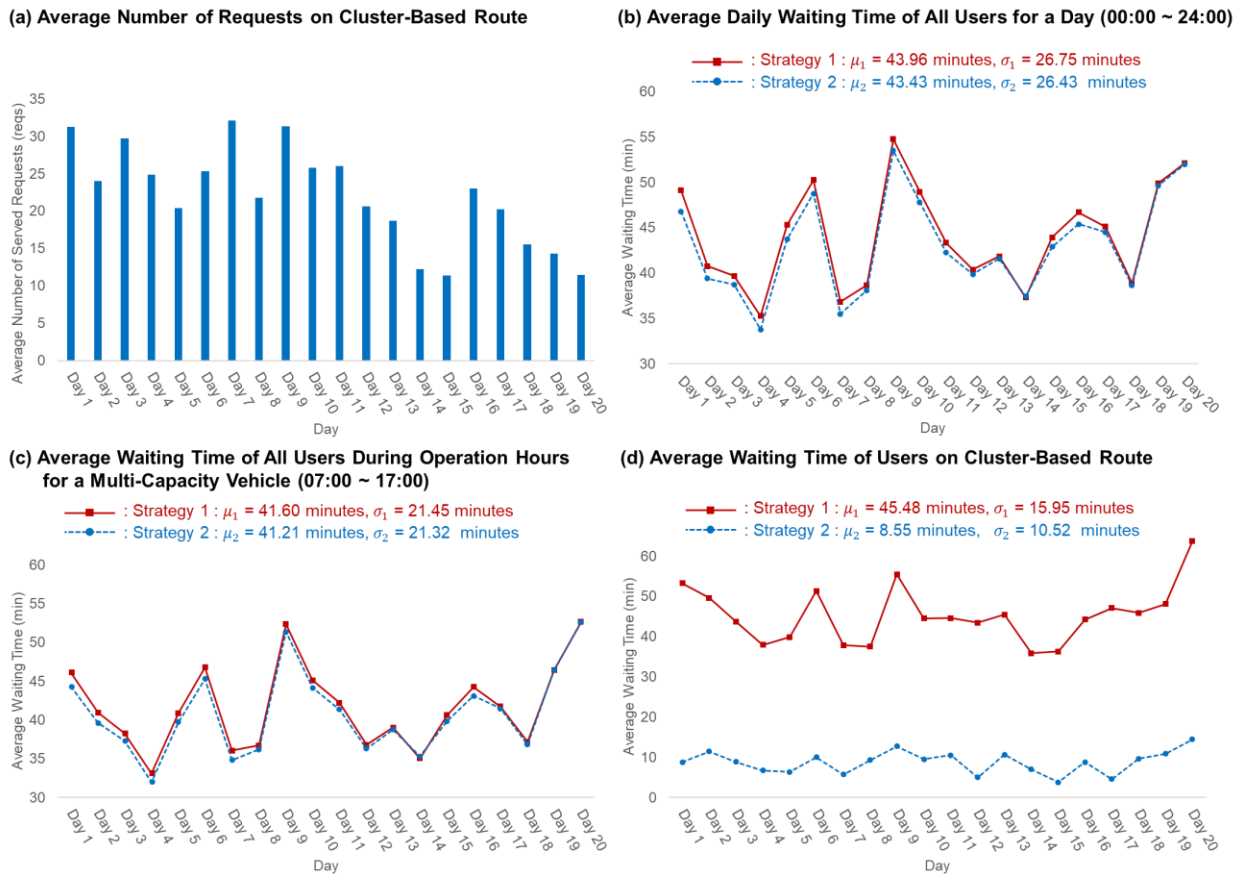

FIGURE 5 Results of Strategy 1 and Strategy 2 in the Large-Scale Network

CONCLUSIONS

Mobility services for the disabled and the elderly have become more crucial due to the worldwide aging phenomenon. The US President's Council of Advisors on Science and Technology presents tailored solutions for the underserved, disabled, and elderly as the objectives of future city infrastructure technologies (23). In large-scale urban areas, many DRT services dispatch vehicles on a first-come-first-serve basis for requests, excluding advance requests. Demand for DRT for the disabled has been steadily increasing; further, recent years have seen a growing need for measures to increase efficiency in high operating costs; thus, efficient DRT services for the disabled are necessary on FCFS.

This study proposed a method to increase the efficiency of a DRT system on FCFS by introducing a multi-capacity vehicle to DRT service for the disabled which operates single-capacity vehicles, and was applied in a toy network and a real large-scale network to verify the method's efficiency. To design a cluster-based route of a multi-capacity vehicle that can service high demands on spatiotemporal similarity, this study proposed a method that consists of four steps. Initial routes were constructed using historical request data, and routes with high spatiotemporal similarity were clustered with a DBSCAN algorithm. A cluster-based route was designed by applying the heuristic algorithm.

The proposed method demonstrated its operability and effectiveness using actual data of DRT service for the disabled in Seoul, Korea. In a toy network, an average waiting time of users on a cluster-based route decreased to 18% of users on current service, and an average waiting time of users on the exact solution of current service decreased to 9% of users on current service. A cluster-based route was designed in a time period with high demand. From the perspective of users on the cluster-based route, the waiting

time decreased dramatically when a multi-capacity vehicle is introduced. A multi-capacity vehicle operated on a cluster-based route served 1.8 times users than an existing single-capacity vehicle. In other words, the result has shown that the average waiting time of users on a cluster-based route greatly improved, and the overall system also improved. This result implied that the proposed method could design a cluster-based route that can introduce a fixed route by effectively measuring spatiotemporal similarity in dispersed demand.

Further research is needed to account for the varying issues by constructing a more efficient cluster-based route via optimizing the cluster-based route and developing a method to determine the optimal number of multi-capacity vehicles. Also, it seems imperative to apply the methodology to real service.

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AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: concept idea and design: Eunhan Ka, Chungwon Lee; data collection: Eunhan Ka; analysis and interpretation of results: Eunhan Ka, Dongju Ka, Chungwon Lee; draft manuscript preparation: Eunhan Ka, Yeonwoo Jung, Chungwon Lee. All authors reviewed the results and approved the final version of the manuscript.

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